

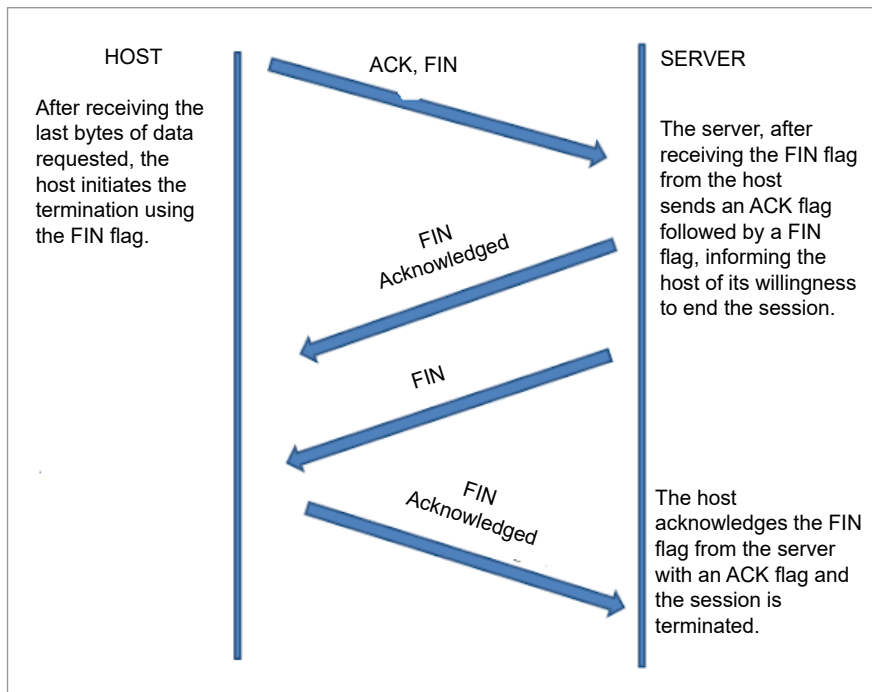


Figure 3: Connection termination (Credits: <https://learningnetwork.cisco.com/s/question/0D53i00000KswSeCAJ/tcp-connection-termination-is-the-diagram-correct>)

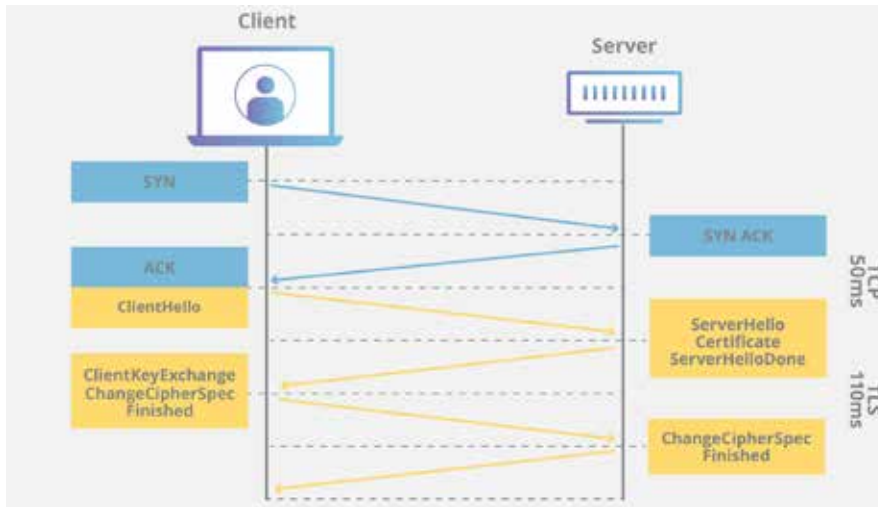


Figure 4: Working of TLS (Credit: <https://www.cloudflare.com/en-in/learning/ssl/what-happens-in-a-tls-handshake>)

- client is interacting with the actual owner of the domain.
- **Server's digital signature:** The server uses its private key to encrypt the client random, the server random, and its DH parameter. This encrypted data functions as the server's digital signature, establishing that the server has the private key that matches with the public key from the SSL certificate matched before.
- **Digital signature confirmed:** The client decrypts the server's digital signature with the public key.

- **Client DH parameter:** The client sends its DH parameter to the server, using the server's public key.
- **Client and server calculate the premaster secret:** The client and server use the DH parameters they exchanged to calculate a matching premaster secret separately.
- **Session keys created:** Now, the client and server calculate session keys from the premaster secret, client random, and server random.
- **Client is ready:** The client sends a "finished" message that is encrypted with a session key.
- **Server is ready:** The server sends a "finished" message encrypted with a session key.
- **Secure symmetric encryption achieved:** The handshake is complete, and communication continues using the session keys.

Using this symmetric key further communication is encrypted and shared. You can find more detailed description about keyless SSL at <https://www.cloudflare.com/en-in/learning/ssl/keyless-ssl/>.

QUIC

QUIC (Quick UDP Internet Connections) is a new, encrypted by default internet transport protocol that has the best qualities of TCP and TLS built on UDP to enhance security with speed. The intended goal is to eventually replace TCL and TLS on the web.

After years of experimentation, the original protocol designed by Google engineers several years ago has now been adopted by the IETF

(Internet Engineering Task Force) for standardisation.

We can see the difference between the handshake time it takes when the connection gets established with normal TCP and TLS vs QUIC in Figure 5.

In the TCP layer we can see a minimum of three round trips, whereas in QUIC it's just one. This is because UDP is connected less and QUIC runs on top of it, making it more secure and faster. UDP is used since it is already established in all the kernels and acts as the thin layer that supports rapid